Breno Livio Silva de Almeida

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Computational biology · Biological sequence representation · Machine learning
brenoslivio@pm.me São Carlos, Brazil
Website Google Scholar 0000-0002-2041-9170 GitHub LinkedIn

Research interests: Representation and prediction of biological sequences, combining handcrafted descriptors, learned representations from foundation models, metagenomic metadata curation and automated machine learning for classification and annotation in microbial ecology.

Education

2024 – Present **PhD in Computer Science — University of São Paulo / Leipzig University.** Double-degree program. Supervisors: André C. P. L. F. de Carvalho and Peter F. Stadler.

2020 – 2024 **BSc in Computer Science — University of São Paulo.** GPA: 9.4/10; rank: 7/116.

Research Experience

07.2025 – 06.2026 **Research Assistant — Helmholtz Centre for Environmental Research (UFZ).** Microbial Data Science group, Department of Computational Biology and Chemistry. Helped develop a machine-learning approach to classify data types across millions of Sequence Read Archive (SRA) runs within EReMiD. Contributed to metagenomic metadata standardisation in related projects and to teaching materials.

07.2022 – 09.2022 **Research Intern — Big Data Biology Lab, Fudan University.** Developed a rarefaction workflow for nearly one billion small open reading frames (smORFs) across 63,402 samples, identifying undersampled soil-associated diversity. Contributed to *A catalog of small proteins from the global microbiome* (*Nature Communications*, 2024).

Research Funding and Fellowships

Awarded approximately **€110,181** in competitive fellowships, research scholarships, and mobility/travel grants across the award periods. Selected awards below; full breakdown in the complete CV.

2025 **Google PhD Fellowship in Health Research** — USD 30,000. Two-year fellowship with Google research mentorship; one of 15 Latin American fellows selected in 2025. Recipients · USP award report.

2025 **ISME Scholar Mobility Fund** — EUR 2,300. Supported a July 2026 research visit to UFZ, Leipzig, through ISME's international microbial-ecology mobility program. Program · USP visit report.

2024 **FAPESP Doctoral Scholarship** — BRL 453,168. *Decoding the language of life: Towards protein classification and design through large language models.* São Paulo Research Foundation (FAPESP); competitive, peer-reviewed; grant #2024/10958-1. FAPESP grant record.

Indicative total at fixed reporting rates: BRL 1 = EUR 0.17; USD 1 = EUR 0.85. These are indicative conversions, not current market rates. Total excludes cash prizes, estimated training values, and consortium budgets; it represents award values, not payments received.

Honors and Awards

2026 **IRCAI Global Top 100 — Outstanding (2025 edition).** Team recognition for BioAutoML from the International Research Centre on Artificial Intelligence (IRCAI), under the auspices of UNESCO. Contributing developer of BioAutoML; currently lead developer of BioAutoML-FAST. Official recognition.

2024 **J. F. Marar Artificial Intelligence Prize for Undergraduate Research.** Winner for *BioDeepFuse: Empowering Researchers in Life Sciences with Deep Learning*; cash prize: **BRL 10,500**. Inaugural ICMC-USP award recognizing academic and social relevance, alignment with the UN Sustainable Development Goals, and academic performance. USP award announcement.

2024 **Prototypes for Humanity.** BioAutoML selected among 100 projects from nearly 3,000 submissions worldwide. International showcase of university innovations addressing social and environmental challenges. Selected project.

Publications

125	77	4	3
Citations (Google Scholar)	Citations (Scopus)	h-index (Google Scholar)	i10-index (Google Scholar)

Google Scholar as of 2026-10-18 · Scopus as of 2026-10-18.

† Equal contribution.

Manuscripts Under Review

- **de Almeida, B. L.**, Bonidia, R. P., Bole, M., Avila-Santos, A. P., Stadler, P. F., Nunes da Rocha, U., & de Carvalho, A. C. P. L. F. (2026). BioAutoML-FAST: an automated machine-learning platform for reusable and benchmarked biological sequence models. *bioRxiv*. **DOI:** 10.64898/2026.04.18.719383. **Under revision at *Bioinformatics* following peer review (major revision).**

Published Journal Articles

- Duan, Y., Santos-Júnior, C. D., Schmidt, T. S., Fullam, A., **de Almeida, B. L.**, Zhu, C., … & Coelho, L. P. (2024). A catalog of small proteins from the global microbiome. Nature Communications, 15(1), 7563. **DOI:** 10.1038/s41467-024-51894-6.
- **de Almeida, B. L.**†, Avila Santos, A. P.†, Bonidia, R. P., Stadler, P. F., Stefanic, P., Mandic-Mulec, I., … & de Carvalho, A. C. (2024). BioDeepfuse: a hybrid deep learning approach with integrated feature extraction techniques for enhanced non-coding RNA classification. RNA Biology, 21(1), 410-421. **DOI:** 10.1080/15476286.2024.2329451.
- Bonidia, R. P., Avila Santos, A. P., **de Almeida, B. L.**, Stadler, P. F., Nunes da Rocha, U., Sanches, D. S., & De Carvalho, A. C. (2022). Information theory for biological sequence classification: A novel feature extraction technique based on Tsallis entropy. Entropy, 24(10), 1398. **DOI:** 10.3390/e24101398.
- Bonidia, R. P., Santos, A. P. A., **de Almeida, B. L.**, Stadler, P. F., da Rocha, U. N., Sanches, D. S., & de Carvalho, A. C. (2022). BioAutoML: automated feature engineering and metalearning to predict noncoding RNAs in bacteria. Briefings in Bioinformatics, 23(4), bbac218. **DOI:** 10.1093/bib/bbac218.

Accepted Journal Articles

- **de Almeida, B. L. S.**†, Bonidia, R.†, Florentino, B.†, Barreto, M. C. A.†, Constantino, J. L. R.†, Tavares da Silva, D.†, Zwicker, G. G., Rocha, U., Goedert, G., Struchiner, C., & de Carvalho, A. AutoAI Hub: From data to society, democratizing machine learning for everyone. Oxford Open Digital Health. **Accepted for publication**.
- Tavares, D., Barreto, M. C. A., **de Almeida, B. L. S.**, Florentino, B. R., Tetzner, F., Goedert, G., Struchiner, C., Bonidia, R., & de Carvalho, A. Dominique: An AI-powered fact-checking chatbot for democratizing access to reliable information. Oxford Open Digital Health. **Accepted for publication**.

Conference Proceedings

- **de Almeida, B. L. S.**†, Queiroz, A. P.†, Santos, A. P. A.†, Bonidia, R. P., da Rocha, U. N., Sanches, D. S., & de Carvalho, A. C. P. D. L. F. (2021, November). Feature importance analysis of non-coding DNA/RNA sequences based on machine learning approaches. In Brazilian Symposium on Bioinformatics (pp. 81-92). Cham: Springer International Publishing. **DOI:** 10.1007/978-3-030-91814-9_8.

Research Software and Resources

BioAutoML-FAST	**Automated machine learning for biological sequences.** Lead developer; expanded sequence types, prediction tasks, and evaluation within the BioAutoML framework. Code · Web application · Preprint.
BioAutoML	**Automated feature engineering and model selection.** Contributing developer. Code · Paper.
BioDeepFuse	**Hybrid deep learning for non-coding RNA classification.** Co-first author of the methods paper. Combines neural networks with handcrafted sequence features. Code · Paper.
MAG2ENA	**Metagenome-assembled genome submission to the European Nucleotide Archive.** Lead developer; developed within NFDI4Microbiota. Code · Web application.

Selected Presentations

2026	**ISME20, Auckland — oral presentation.** Protein language models for protein-family annotation in environmental metagenomes. Program MS13.
2026	**VAAM, Berlin — oral presentation.** Protein language models for functional annotation. Abstract SL-CM-01.
2025	**BAGECO, Graz — flash talk; Helmholtz AI, Messe Karlsruhe — oral presentation.** Protein language models for uncharacterized microbial proteins. BAGECO abstract · Helmholtz AI.

Teaching and Mentoring

2025 – 2026	**Instructor — UFZ Summer School**, Leipzig. Unix/Bash/Git, machine learning, and alignment-based and alignment-free annotation.
06.2025	**Co-organizer and instructor — BAGECO metagenomics workshop**, Graz.
05.2021 – 11.2022	**Teaching Assistant — University of São Paulo.** Four undergraduate courses.

Academic Service

Journal reviewer: *Briefings in Bioinformatics* and *Journal of Thermal Biology*. Member of six undergraduate thesis examination committees.

Technical Skills

Methods: sequence representation, automated machine learning, metagenomic sequence analysis. **Tools:** Python, R, scikit-learn, TensorFlow, Git, Linux. **Sequence analysis:** Biopython, BLAST, Prodigal, OrthoFinder, Anvi'o, HMMER, Infernal, DIAMOND, MMSeqs2. **Languages:** Portuguese (native), English (C1), German (A1.2).